

Assignment 06 - SQL Views

Introduction

A SQL View is a virtual table based on the result-set of an SQL statement. Views simplify complex queries, enforce consistent logic, and help control what data users can see. Below explains when to use Views and compares Views with Functions and Stored Procedures.

When To Use a SQL View

You can use a view when you want a reusable, user-friendly representation of data without rewriting SQL statements. Views are used to simplify complex SELECT queries, enforce read-only access to certain fields, abstract underlying schema changes and create structured reporting queries.

Views, Functions, and Stored Procedures

Views

As described above, a SQL View is a virtual table created from a query. It does not store physical data (unless materialized), but it simplifies reading complex queries. It is not a good solution for business logic calculations, updating multiple tables or for loops, conditions, or procedural logic.

Functions

A Function returns a single value or a table. It is useful for reusable calculations or filtering logic. Unlike views, functions can use parameters to change the results of a query.

Stored Procedures

A Stored Procedure contains business logic, multiple SQL commands, and can perform read/write operations. It can be used for business logic, logging, auditing, or modifying multiple tables, returning multiple result sets, and ETL, batch processing, or API integrations. Like Views or Functions, Stored Procedures (Sprocs or Procs) are a Named Set of SQL Statements.

Conclusion

Views are best for simplifying and reusing SELECT logic and controlling data visibility. Functions add parameterized, query-usable logic; stored procedures are for procedural tasks, data changes, and complex workflows. Each is used based on whether you need a virtual table (View), a query-friendly reusable function (Function), or an executable task with side effects (Stored Procedure).